Relative Covariance

Nuno Carvalho

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This writeup derives the covariance in phenotype (and its genetic components) between pairs of relatives, and how it relates to relatedness, heritability, and the variance components of a trait. It is currently a stub.

1 Setup

Introduce the phenotype model and notation for a pair of individuals j and k, reusing the additive model from the "Heritability" entry ($\mathbf{y} = \mathbf{g} + \boldsymbol{\epsilon}$, with $\mathbf{g} \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{G})$). Define relatedness/kinship between the pair.

2 Covariance between relatives

Derive $\mathrm{Cov}[y_j, y_k]$ in terms of the relatedness G_{jk} and the additive genetic variance σ_g^2, and show how the phenotypic correlation between relatives relates to heritability h^2.

2.1 Special cases

Work through familiar relationships (e.g. parent–offspring, full siblings, monozygotic vs. dizygotic twins) as instances of the general result.

3 Beyond the additive model

Comment on extensions: dominance/epistatic contributions, shared environment, and assortative mating, and how each perturbs the relative covariance.

References